

10 · NETWORK FLOW

$0 \leq f(e) \leq u(e)$ capacity
in = out at every node but s,t conservation
 $val(f) = \sum_j f(s,j)$

Cut $S = [X, V \setminus X]$, $s \in X$, $t \notin X$:

$cap(S) = \sum u(i,j)$ $i \in X, j \notin X$ FORWARD ARCS ONLY

Backward arcs can't carry anything that way, so they impose no limit; counting them inflates the cut.

$val(f) \leq cap(S)$ every flow, every cut weak
 $\max val(f) = \min cap(S)$ strong

Find a flow and a cut with the same number $\implies$ both optimal. One line.

residual: FORWARD (i,j): $u(e) - f(e)$ spare
BACKWARD (j,i): $f(e)$ flow to UNDO
drop residual capacity 0

Ford–Fulkerson
1 build residual 2 ANY s–t path (none $\rightarrow$ MAX)
3 $k = \min$ residual on path 4 $f \pm k$ fwd/bwd

min cut free: $X =$ nodes reachable from s in the FINAL residual. Check $cap(S) = val(f)$.

Edmonds–Karp = always a shortest path (BFS). FF: $O(|E| \cdot U)$ pseudopoly · EK: $O(V \cdot E^2)$ poly.

Transformations: many sources/sinks $\rightarrow$ super-source/sink · node capacity $\rightarrow$ split $v_{in} \rightarrow v_{out}$ with that arc · undirected edge $\rightarrow$ two opposite arcs.

11 · TSP

EULER = every EDGE once (easy). HAMILTON = every NODE once (NP-complete). TSP = min-weight Hamiltonian cycle.

symmetric $c_{ij} = c_{ji}$ $(n-1)!/2$ tours
asymmetric $c_{ij} \neq c_{ji}$ $(n-1)!$ tours
metric $c_{ik} \leq c_{ij} + c_{jk}$ -ratios need it

degree: $\sum_{i \neq j} x_{ij} = 1 \forall j$; $\sum_{j \neq i} x_{ij} = 1 \forall i$

Degree constraints alone = the assignment problem $\rightarrow$ they permit subtours.

	constraint	count	relax
SEC	$\sum_{\{i,j \in U\}} x_{ij} \leq U -1$, $2 \leq U \leq n-1$	exp	tight
MTZ	$u_i - u_{j+1} \leq (n-1)(1-x_{ij})$, $u_1 = 1, 2 \leq u_i \leq n$	$O(n^2)$	weak

Use the lecture's $(n-1)$, not the textbook's n .

	ratio	how degrees are fixed
MST-doubling	2	double every edge
Christofides	3/2	min-weight matching on ODD-degree vertices
nearest nb.	none	—

Both: MST $\rightarrow$ make degrees even $\rightarrow$ Euler tour $\rightarrow$ shortcut. The papers mislabel MST-doubling as "Christofides" — execute what's described, add one line noting the correct names.

► COUNTEREXAMPLE — 7x in 3 papers

Asked in matroids · flow · TSP · convexity. Always the same four moves:

1 SMALLEST instance meeting EVERY stated assumption — check them explicitly
2 what the claim predicts
3 what actually happens
4 "therefore the statement is false in general"

Levers that worked: matroid — $A, B \in I$, $|A| < |B|$, nothing in $B \setminus A$ extends A · heuristic — greedy commits to a cheap early edge · convexity — non-convex K : $\min x^2$ s.t. $x^2 \geq 1 \rightarrow x = \pm 1$, min not unique.

The flow graph that breaks most claims (SS25 P6, used for BOTH parts):

s→a=1, s→b=1, a→c=1, b→c=1, c→t=3
max flow 2 (EVEN, though all caps odd)
min cut ({s},rest)=2; add $\lambda=2$ to every cap
 $\rightarrow$ ({t},rest)=5 beats the old cut's 6 $\implies$ min cut MOVED

Classify: pair one-to-one=assignment · one budget max value=knapsack · min #containers=bin packing · cover every element=set covering · visit every node=TSP · pick nodes so every edge touched=vertex cover.

12 · NONLINEAR, UNCONSTRAINED

$\nabla f = 0$ $\rightarrow$ critical point (saddles too)
 $H_f = (\partial^2 f / \partial x_i \partial x_j)$ always symmetric

Solve $\nabla f = 0$ for all points — factor, don't divide ($x \cdot (\dots) = 0$ has two branches). Substitute each point separately.

det	tr	point is
>0	>0	pos. def $\rightarrow$ MIN
>0	<0	neg. def $\rightarrow$ MAX
<0	—	indefinite $\rightarrow$ SADDLE
$=0$	—	$\rightarrow$ eigenvalues

Minors D_{-k} : all $>0 \rightarrow$ min · alternating starting negative ($D_1 < 0, D_2 > 0, \dots$) $\rightarrow$ max · none zero but neither pattern $\rightarrow$ saddle · any $D_{-k} = 0 \rightarrow$ test INAPPLICABLE.

Eigenvalues $\det(H - \lambda I) = 0$: all $+ \rightarrow$ min, all $- \rightarrow$ max, mixed $\rightarrow$ saddle, some 0 $\rightarrow$ inconclusive.

A zero minor does NOT mean semidefinite — the test can't decide. $H = \begin{bmatrix} 0 & \alpha \\ \alpha & 0 \end{bmatrix} \rightarrow \lambda = \pm \alpha \rightarrow$ indefinite $\rightarrow$ saddle.

Do FIRST: if $H > 0$ everywhere $\implies$ strictly convex $\implies$ no maxima exist, any critical point is the unique global min. ($H > 0 \implies$ strictly convex, not conversely $-x^4$.)

13 · CONVEXITY

convex objective + convex feasible set $\implies$ every LOCAL min is GLOBAL.

SET $\lambda x + (1-\lambda)y \in C \quad \forall x,y \in C, \lambda \in [0,1]$
"the whole segment stays inside"
FUNCTION $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$
"the chord lies ABOVE the graph"

Strictly convex = $<$ for $x \neq y$. x^2 yes; $|x|$ convex but not strictly. Concave = reverse. Affine $a^T x + b$ = both (chord IS the graph).

	compact	convex
circle curve $x^2 + y^2 = r^2$	✓	✗
closed disk $\leq r^2$	✓	✓
halfplane	✗	✓
intersection of convex		✓ always
union of convex		✗ generally

Open / closed / bounded / compact

Open = every point has a ball inside (strict $<$) · Closed = contains its boundary (non-strict $\leq$; complement open) · Bounded = fits in a finite box · COMPACT = closed AND bounded (Heine–Borel).

"Not open" $\neq$ "closed": $[0, 1]$ is neither; $\mathbb{R}^n$ is both. $(0, 1)$: max never attained — not closed. $[0, \infty)$: runs away — not bounded.

Weierstrass: non-empty compact + continuous $f \implies$ global min AND max are attained.

Proving convexity

definition f abstract/non-differentiable
Hessian $H \geq 0$ on a CONVEX domain
composition built from known pieces
counterex. one triple $(x,y,\lambda) \rightarrow$ DISprove

$H \geq 0$ everywhere $\iff$ convex. $H > 0$ everywhere $\implies$ strictly convex (converse false: x^4).

keeps: αf ($\alpha \geq 0$) · $f+g$ · $f(Ax+b)$ · $\max\{f_1..f_m\}$ · $h \circ g$ (h convex, non-decreasing)

Norm proof — say every line

$\|\lambda x + (1-\lambda)y\| \leq \|\lambda x\| + \|(1-\lambda)y\|$ triangle ineq.
 $= |\lambda| \|x\| + |1-\lambda| \|y\|$ homogeneity
 $= \lambda \|x\| + (1-\lambda) \|y\|$ $\lambda \in [0,1] \implies$ both ≥ 0
 $\leftarrow$ SAY THIS LINE

14 · KKT

min $f(x)$ s.t. $g_i(x) \leq 0$, $h_j(x) = 0$
 $g \geq 0 \rightarrow -g \leq 0$ max $f \rightarrow$ min $-f$
ACTIVE $g_i(x^*) = 0$ INACTIVE $g_i(x^*) < 0$

$L = f + \sum \lambda_i g_i + \sum \mu_j h_j$ (MIN)
 $\lambda_i \geq 0$ SIGN-CONSTRAINED μ_j FREE

Violating $g_i \leq 0$ must increase the objective you minimise $\implies \lambda \geq 0$. An equality breaks either way $\implies \mu$ free. Same SOB pattern as §4. λ is a shadow price.

stationarity	$\nabla f + \sum \lambda_i \nabla g_i + \sum \mu_j \nabla h_j = 0$
primal feas.	$g_i \leq 0, h_j = 0$
dual feas.	$\lambda_i \geq 0$
compl. slack.	$\lambda_i \cdot g_i(x^*) = 0$

case split, per i: $\lambda_i = 0$ (inactive)
or $g_i(x^*) = 0$ (active)
DISCARD, SAY WHY: $\lambda_i < 0 \rightarrow$ dual infeasible
 $g_i > 0 \rightarrow$ primal infeasible

Kill cases early from the problem statement: two distinct points exist $\implies r > 0 \implies \mu r = 0 \implies \mu = 0$.

	requires	gives
Slater	f, g_i convex; h_j affine; an $\bar{x}$ with every $g_i(\bar{x}) < 0$ strictly	KKT point IS a global optimum — stop
LICQ	active ∇g_i + all ∇h_j linearly independent	candidates only — evaluate f and compare

Verify Slater by exhibiting the point — centroid with radius +1, so every $g_i \leq -1 < 0$. The +1 manufactures strictness. Only equality constraints $\implies$ Slater always fails (no strict version).

MC: "a KKT point is always a local optimum" = FALSE. "for a convex problem a KKT point is a global minimiser if Slater holds" = TRUE.

RECIPE
1 EXISTS compact + continuous $\implies$ min,max
2 CQ Slater? else LICQ
3 SOLVE $L \rightarrow$ 4 blocks $\rightarrow$ case split $\rightarrow$ discard
4 COMPARE f at survivors — SKIP IF SLATER HELD